Floor beam B1

Simply supported floor beam · Example building, axis B

logo

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Simply supported beam, uniformly loaded, laterally restrained by the slab. Profile IPE 300, steel S355.

Geometry and loads

$L = 6.0\,\mathrm{m}$ *span*

$q_{Ed} = 25\,\mathrm{kN/m}$ *design load, ULS 6.10b*

$q_{ser} = 18\,\mathrm{kN/m}$ *characteristic load, SLS*

$$M_{Ed} = \frac{q_{Ed}\,L^2}{8} = \frac{25\,\mathrm{kN/m}\cdot(6.0\,\mathrm{m})^2}{8} = 112.5\,\mathrm{kN{\cdot}m}$$

$$V_{Ed} = \frac{q_{Ed}\,L}{2} = \frac{25\,\mathrm{kN/m}\cdot 6.0\,\mathrm{m}}{2} = 75\,\mathrm{kN}$$

Section and material

IPE 300, values from the profile table.

$h = 300\,\mathrm{mm}$

$b = 150\,\mathrm{mm}$

$t_w = 7.1\,\mathrm{mm}$

$t_f = 10.7\,\mathrm{mm}$

$r = 15\,\mathrm{mm}$

$A = 5381\,\mathrm{mm^2}$

$A_v = 2568\,\mathrm{mm^2}$ *shear area, z-z*

$I_y = 8356\cdot 10^4\,\mathrm{mm^4}$

$W_{el,y} = 557.1\cdot 10^3\,\mathrm{mm^3}$

$W_{pl,y} = 628.4\cdot 10^3\,\mathrm{mm^3}$

$f_y = 355\,\mathrm{MPa}$

$E = 210\,\mathrm{GPa}$

$\gamma_{M0} = 1.0$

$\gamma_{M1} = 1.0$

Classification

Outstand flange in compression, Table 5.2.

$$\varepsilon = \sqrt{\frac{235\,\mathrm{MPa}}{f_y}} = \sqrt{\frac{235\,\mathrm{MPa}}{355\,\mathrm{MPa}}} = 0.8136$$

$$c_f = \frac{b - t_w - 2r}{2} = \frac{150\,\mathrm{mm} - 7.1\,\mathrm{mm} - 2 \cdot 15\,\mathrm{mm}}{2} = 56.45\,\mathrm{mm}$$

$$\frac{c_f}{t_f} = \frac{56.45\,\mathrm{mm}}{10.7\,\mathrm{mm}} = 5.276$$

$$\frac{c_f}{t_f} = 5.276 \le 9\varepsilon = 7.322$$

> $cls = 1$

The flange is class 1, so the plastic section modulus is used.

Bending resistance, 6.2.5

$$M_{c,Rd} = \frac{W_{pl,y}\, f_y}{\gamma_{M0}} = \frac{628.4 \cdot 10^3\,\mathrm{mm}^3 \cdot 355\,\mathrm{MPa}}{1.0} = 223.1\,\mathrm{kN{\cdot}m} \qquad (6.13)$$

$M_{Ed} = 112.5\,\mathrm{kN{\cdot}m} \le M_{c,Rd} = 223.1\,\mathrm{kN{\cdot}m}$ ✓ OK η = 0.50

Shear resistance, 6.2.6

$$V_{pl,Rd} = \frac{A_v\, f_y}{\sqrt{3}\,\gamma_{M0}} = \frac{2568\,\mathrm{mm}^2 \cdot 355\,\mathrm{MPa}}{\sqrt{3} \cdot 1.0} = 526.3\,\mathrm{kN} \qquad (6.18)$$

$V_{Ed} = 75\,\mathrm{kN} \le V_{pl,Rd} = 526.3\,\mathrm{kN}$ ✓ OK η = 0.14

$V_{Ed} = 75\,\mathrm{kN} \le 0.5\,V_{pl,Rd} = 263.2\,\mathrm{kN}$ ✓ OK η = 0.28 *no M-V interaction, 6.2.8(2)*

Deflection

Limit L/250 for the characteristic combination.

$$\delta = \frac{5\,q_{ser}\,L^4}{384\,E\,I_y} = \frac{5 \cdot 18\,\mathrm{kN/m} \cdot (6.0\,\mathrm{m})^4}{384 \cdot 210\,\mathrm{GPa} \cdot 8356 \cdot 10^4\,\mathrm{mm}^4} = 17.31\,\mathrm{mm}$$

$$\delta_{lim} = \frac{L}{250} = \frac{6.0\,\mathrm{m}}{250} = 24\,\mathrm{mm}$$

$\delta = 17.31\,\mathrm{mm} \le \delta_{lim} = 24\,\mathrm{mm}$ ✓ OK η = 0.72

Summary

Bending utilization 0.5043, shear 0.1425, deflection 0.7213. IPE 300 S355 is adequate.